Water Crisis Decision Navigator

Expert Analysis of the Project Across Nine Business and Technical Questions

Reference date: 25 October 2026. **Initial deployment geography:** Bulgaria. **Potential expansion market:** European Union.

Executive Expert Assessment

The project has commercial potential. Its primary value, however, lies less in satellite imagery or artificial intelligence in isolation than in converting environmental observations into justified, accountable management action.

The market for water monitoring and flood intelligence already includes strong technology providers. Water Crisis Decision Navigator should therefore be positioned as a specialised decision-support system for public authorities, water utilities and businesses.

Proposed operating sequence: Observation → anomaly assessment → identification of the competent authority → legal verification → expert approval → implementation monitoring.

These stages must be distinguished carefully:

- **Anomaly detection:** the available data reveal an unusual change in water parameters.
- **Risk assessment:** the system explains who or what might be affected.
- **Establishment of a violation:** this requires the applicable legal criteria, reliable measurements and, where required, a decision by the competent authority.
- **Response:** the user receives a proposed action sequence, assigns responsibilities and monitors the outcome.

The last two stages must remain under the supervision of authorised professionals.

Why This Project Matters Now

€822 billion: estimated losses in the EU from weather- and climate-related extremes over 1980–2024, expressed in 2024 prices, according to the European Environment Agency (EEA).

47%: the share of hydrological hazards in those losses, predominantly flooding.

According to the EEA, the years 2021–2024 accounted for more than €208 billion of the cumulative losses in the period examined. These numbers establish the scale of the policy problem; they do not mean that the same sums constitute the addressable market for our software.

Bulgaria offers particularly relevant evidence:

1. On 26 September 2025, Bulgaria's Ministry of Environment and Water urgently requested data on alternative water sources, relevant permits and water-abstraction volumes from water-supply organisations to respond to a water crisis.

2. On 9 October 2025, the ministry instructed the four River Basin Directorates to prepare combined maps of areas at significant flood risk, based on cadastral, satellite and other specialised information. An initial pilot to integrate spatial data was planned for the Black Sea area, in the Varna directorate.

3. On 14 October 2026, the European Commission announced the Water Resilience Investment Accelerator to support 20 innovative water-sector pilot projects. The Commission estimated the investment gap for implementing existing EU water legislation at roughly €23 billion per year.

The implication is twofold. Demand for integrating water data and administrative decisions is real; nevertheless, Bulgaria is already developing digital infrastructure. Our product must integrate with existing solutions rather than assume none exist.

1. Potential Customers

1.1. Core Customer Groups

Customer segment	Example potential buyer	Primary use case	Priority
River basin authorities	Bulgaria's River Basin Directorates	Water-status oversight, jurisdiction and response coordination	Very high
Environmental inspectorates	Regional Inspectorates of Environment and Water	Verifying alerts, organising sampling and tracking regulatory measures	Very high
Water-supply operators	Sofiyska Voda and regional water and sewerage utilities	Water quality, incidents, alternative sources and public-authority coordination	High
Municipalities and emergency headquarters	Municipal civil-protection authorities	Floods, evacuation and protection of infrastructure	Medium
Industrial enterprises	Aurubis Bulgaria; Solvay Sodi	Water abstraction, discharges, incidents and environmental risks	High
Insurers	DZI and other property insurers	Exposure assessment, loss prevention and portfolio management	Medium
Engineering and environmental consultancies	Monitoring, environmental assessment and industrial-safety firms	Additional client-facing analytical capability	Complementary channel

According to Bulgaria's Ministry of Environment and Water, the country's water-governance framework includes the ministry, four River Basin Directorates and 16 Regional Inspectorates of Environment and Water, each with its respective geographic and institutional remit. See the ministry's official website: https://www.moew.government.bg/.

1.2. Particularly Promising Bulgarian Institutions

River Basin Directorate	Administrative centre	Candidate pilot
Danube Region	Pleven	River pollution and cross-border cooperation
Black Sea Region	Varna	Coastal-water quality, inundation and runoff
East Aegean Region	Plovdiv	River pollution and industrial water use
West Aegean Region	Blagoevgrad	Floods, river-water quality and interagency action

The ministry confirms the directorates and their locations. My priority would be one River Basin Directorate as an expert design partner, combined with one water utility or industrial operator as a potential paying customer. That gives us two fundamentally different tests: whether the proposed procedure satisfies public-administration requirements, and whether an organisation will pay for the resulting operational value.

1.3. Specific Companies Worth Interviewing

Sofiyska Voda, the water supplier for Sofia, already uses laboratory and operational monitoring, including water-quality parameters and management systems. The relevant value proposition is thus better integration and crisis-response procedures—not the claim that we are introducing monitoring for the first time. Source: https://www.sofiyskavoda.bg/.

Aurubis Bulgaria, Pirdop operates an industrial facility with wastewater, stormwater and drainage-water treatment. Its published environmental investments and treatment systems make it a particularly relevant research candidate for interviews about environmental risk and incident response. Source: https://www.aurubis.com/.

Solvay Sodi, Devnya is another industrial candidate, given its regulated activity and operational environmental risks. Potential relevance does not imply willingness to purchase; this must be tested in customer interviews.

DZI offers commercial property insurance that can include natural-catastrophe and business-interruption protection. This demonstrates that insurance products exist for which water-risk intelligence might be useful.

Important qualification: these are potential market-research subjects. We have no confirmation that any organisation wishes to purchase our platform or has agreed to participate in a pilot.

1.4. Who Should We Approach First?

1. **River Basin Directorate:** validate institutional responsibilities, applicable law and actual procedures.
2. **Water utility:** identify recurring operational tasks and test a first commercial pilot.
3. **Industrial enterprise:** estimate avoided administrative and incident-related costs.
4. **Municipality:** develop a later-stage flood-response scenario.
5. **Insurance company:** approach once reliable flood-risk models can be integrated.

Insurers may have substantial purchasing power, but they need especially accurate risk models and often already use established vendors.

2. Customer Problems and Pain Points

Stavri Nikolov rightly stressed the importance of investigating customer problems. We should not assume that public authorities simply lack data: often there are many datasets, but they are not used effectively together.

2.1. Principal Problems

Problem	What happens in practice	Potential product value
Fragmented information	Satellite images, sensors, laboratory records and permit registers sit in separate systems	One consolidated incident record
Inability to verify anomalies immediately	Changes in water colour or turbidity may have natural causes	Confidence assessment and advice on verification
Unclear institutional responsibility	Staff must identify which authority has the legal power to act	Jurisdiction and responsibility matrix
Complex legal interpretation	Users must find current EU rules, national implementation and specific permits	Verifiable legal knowledge base
Poor coordination	Information moves between institutions by letters, email and telephone	Task assignments and status tracking

Problem	What happens in practice	Potential product value
Late consequence assessment	Teams act without a clear picture of potentially affected assets	Threat-development scenarios
Limited audit trail	Later it can be hard to establish who received an alert and what was done	Chronology of decisions and evidence
Excessive false alarms	Experts devote time to alerts that do not require intervention	Prioritisation by confidence and consequences

2.2. The Inefficient Process We Should Investigate

A typical interview hypothesis is as follows: an officer receives a report of possible pollution, checks a map manually, searches monitoring records, contacts a laboratory, identifies downstream water users, establishes the responsible institution, looks for the relevant legal provision and prepares an official notification. The platform should reduce manual work across this sequence without replacing legally required verification and decisions.

2.3. What We Do Not Yet Know

Before asserting a proven customer pain point, we need answers to these questions:

- How long does it take to handle one alert?
- How many alerts occur each month?
- Where do delays usually arise?
- What proportion of alerts is ultimately unconfirmed?
- How do staff identify the correct recipients of notices?
- Which systems are already in use?
- Which decisions cannot lawfully be automated?
- What happens if the response is late?

Until interviews have been conducted, these remain research hypotheses, not proven deficiencies at any named institution.

3. Decision-Makers and Budget Holders

In public administration, we must separate three roles: **end user**, who works with the system; **implementation decision-maker**, who identifies the organisational need; and **budget owner/procuring authority**, which can commit the funds. These functions may belong to different people or departments.

Organisation	End user	Initiator or decision-maker	Possible payer
River Basin Directorate	Environmental officer, hydrologist, inspector	Directorate leadership and specialist units	Authorised public budget holder or funded project
Regional environmental inspectorate	Inspector and analyst	Inspectorate management	Relevant public institution or programme
Municipality	Emergency headquarters, GIS specialist	Mayor, deputy mayor or emergency-service head	Municipal budget or earmarked funding
Water utility	Dispatcher, laboratory, environmental specialist	Technical or operations director	Utility/operator

Organisation	End user	Initiator or decision-maker	Possible payer
Industrial site	Environmental/EHS specialist, engineer	Plant director or EHS director	Company
Insurer	Underwriter, risk analyst	Property insurance head or chief risk officer	Insurance company

3.1. How to Present the Offer

Public sector: "We help your experts determine necessary actions faster when a water-related threat appears and document compliance with established procedures."

Businesses: "We help detect operational risks earlier and launch verified measures to protect your facilities more quickly."

Insurers: "We combine hazard information with insured-asset exposure to accelerate portfolio analysis and risk-management action."

3.2. Additional Parties Influencing Procurement

Approval from the environmental team is not enough. An implementation may depend on IT, cybersecurity, legal, finance, data protection, procurement and existing GIS-vendor contracts. EU public bodies must also comply with their applicable national public-procurement rules.

The important interview question is therefore not only "Would the system help you?" but also "Who could approve a paid pilot?"

4. Competitor Analysis

4.1. Four Established Solutions

Solution	Verified publicly described functions	Strength	Relationship to our concept
Copernicus Emergency Management Service	Flood forecasting, satellite-based monitoring, crisis and damage mapping	Scale, official EU infrastructure and accessible open datasets	Input-data provider and alternative monitoring capability
FloodMapp	Predicted flood depths and footprints, frequent updates, impact on roads, properties and infrastructure	Detailed operational flood intelligence	Direct competitor in flood-response use cases
Fathom	Global flood hazard maps, individual-asset and portfolio assessment, insurance risk models, APIs	Mature insurance analytics	Strong competitor in the insurance segment
EOMAP eoapp AQUA	Satellite-based turbidity, chlorophyll, algal indicators and surface-temperature monitoring, historic trends and threshold alerts	Direct specialisation in water quality	Closest competitor for a pollution-monitoring scenario

Primary sources: https://emergency.copernicus.eu/about/ ; https://www.floodmapp.com/ ; https://www.fathom.global/sector/insurance/ ; https://eomap.com/start-download/eoapp-aqua-leaflet.

4.2. Individual Competitor Assessments

Copernicus EMS. Its services include the European Flood Awareness System (EFAS), the Global Flood Awareness System (GloFAS), and mapping products activated through authorised users. GloFAS offers global river forecasts, while national and local services remain essential for detailed response. Building a competing satellite dataset from scratch would be impractical. We should treat eligible Copernicus products as input sources.

FloodMapp. It already supports the identification of roads and properties at risk, evacuation planning, resource deployment and impact documentation. The NowCast offering can update flood maps approximately hourly. We must not tell a jury that all rivals merely display maps without informing decisions. The additional layer we could provide is a verified, country-specific regulatory procedure, an interagency responsibilities model and execution tracking.

Fathom. A major provider for insurers, with maps, asset- and portfolio-level assessments, pricing support and catastrophe-loss models. A flood-risk map alone is not a defensible offering to this segment. We need to integrate reliable external analytics and concentrate on the further step: how to organise prevention for the exposed assets.

EOMAP eoapp AQUA. Especially relevant because it already links satellite-derived indicators such as turbidity, chlorophyll and algae to time series, custom thresholds and notifications. We cannot claim to be the first to combine satellite-based water-quality observation with alerting.

4.3. An Additional Important Competitor: Esri ArcGIS Workflow Manager

ArcGIS Workflow Manager already supports task sequences, work assignments, role-based activities and notifications. Accordingly, the combination “map + tasks + notifications” is not unique either. Documentation: https://doc.esri.com/en/arcgis-workflow-manager/latest/help/an-introduction-to-arcgis-workflow-manager.html.

Our more precise differentiator is a **specialised platform that connects a specific water hazard with verifiable applicable legal duties, authorities' jurisdiction, required procedures and traceable implementation**.

4.4. Functional Comparison

This comparison reflects publicly described functionality rather than a complete technical audit. **“Not established” does not mean a vendor definitely lacks a feature.**

Capability	Copernicus	FloodMapp	Fathom	EOMAP	Target navigator
Flood monitoring or modelling	Yes	Yes	Yes	Not a principal focus	Through external datasets
Satellite water-quality analysis	Not a principal CEMS function	Not established	Not established	Yes	Through integration
Water-hazard notifications	Selected services	Yes	Product-dependent	Yes	Planned
Affected-asset assessment	Selected products	Yes	Yes	Partly	Planned
Mapping of national institutional powers	Not established as a standard feature	Not established	Not established	Not established	Key hypothesis
Links to applicable rules and versions	Not established as a standard feature	Not established	Not established	Not established	Key hypothesis
Expert-approved step-by-step procedure	Some operational support	Some operational support	Not a main function	Not established	Key hypothesis
Execution tracking and decision log	Depends on local setup	Depends on integration	Depends on integration	Not established	Planned

4.5. Recommended Competitive Strategy

Do not try to create a universal satellite-analysis system from scratch. Use a modular architecture:

- **Copernicus and other admissible sources** for hydro-meteorological and mapping data.
- **Laboratories and sensors** to confirm relevant water parameters.
- **Verified national legal sources** to identify procedures that may apply.
- **Our own review and workflow layer** to coordinate people, approvals and subsequent evidence.

Water Crisis Decision Navigator can potentially complement existing government and commercial systems instead of displacing them.

5. Competitive Advantage and Genuine Innovation

5.1. The Innovation That Can Matter

In my assessment, the main innovation is not any individual AI model. It is a domain-specific architecture for actionable water-governance decisions, bringing together six layers:

Component	Function	Customer benefit
1. Environmental Intelligence	Satellite, sensor, laboratory and hydrological-model data	Understand the condition of the water body
2. Risk Intelligence	Hazard detection, confidence rating and potential consequences	Prioritise intervention
3. Institutional Intelligence	Geographic jurisdiction, competent institutions and legal powers	Reduce coordination time
4. Legal Intelligence	Applicable EU law, national legislation, permits and procedures	Legally justified action
5. Decision Workflow	Draft steps, task owners, approvals, deadlines and notifications	Organise the response
6. Accountability	Data lineage, versions, decisions, notifications and acknowledgements	Auditability and accountability

The first two layers are already strongly represented in the market. The proposed combination of layers 3–6 is our central commercial hypothesis.

5.2. What the User Should See

Imagine selecting a river segment where satellite observations show a sudden increase in turbidity. Opening the incident should reveal four linked panels.

Observation panel: water body and location; image acquisition time; data source; anomaly type and magnitude; historical observations; confidence level; whether laboratory confirmation exists.

Potential-impact panel: possible affected area; water-intake points and other sensitive assets; businesses and settlements in the relevant area; whether the situation could worsen; limitations and uncertainty in any forecast.

Legal and institutional panel: relevant River Basin Directorate; potentially competent environmental inspectorate; applicable instruments; permit conditions when available; notice and verification requirements; hyperlinks to exact provisions.

Action panel: inspect the underlying data; initiate targeted sampling when warranted; prepare a notice to the competent authority; assess potential effects on water abstraction; record decisions; monitor tasks until the incident is closed.

Each action should display one of the following statuses: **system-suggested; approved by specialist; in progress; completed; rejected with justification**.

5.3. Critical Scientific and Technical Limitation

Satellite imagery cannot reliably identify every pollutant. Optical remote sensing can help estimate turbidity, chlorophyll and selected other indicators. A change in water colour alone, however, does not automatically establish:

- Nitrate concentration.
- Presence of a specific toxic compound.
- Source of the pollution.
- Identity or responsibility of an industrial operator.
- Exceedance of a legally applicable limit.

Developing and validating satellite-based water-quality models requires in-situ measurements, as US Geological Survey guidance on remote-sensing verification stresses.

The interface must explicitly distinguish **Anomaly**, **Verification required**, and **Confirmed incident**. Not every anomaly becomes a confirmed case of pollution.

5.4. Clear Positioning for the Jury

Water Crisis Decision Navigator transforms environmental observations into source-verified, jurisdiction-specific response workflows, helping responsible organisations determine what to check, who to involve, what to do next, and how to document their decisions.

That is more defensible than claiming to have no competitors.

5.5. Long-Term Differentiation

Established vendors could reproduce a similar interface. A durable advantage should therefore rely on assets that cannot be created overnight:

1. An expert-verified register of national water-protection procedures.
2. Jurisdiction and competence matrices covering multiple authorities.
3. A library of tested emergency scenarios.
4. Integrations with real organisations' operational systems.
5. A record of expert-reviewed decisions and subsequent outcomes.
6. A replicable method for adapting the platform to another EU Member State.

Together, these could form a specialist knowledge and procedural infrastructure, not just another mapping application.

6. User Stories and Operational Scenarios

Five scenarios are recommended. They can ultimately be independent modules, but the first MVP should implement **one** complete use case.

Scenario 1. Suspected River Pollution After Heavy Rain

MVP priority: 1. User: River Basin Directorate or regional environmental-inspectorate specialist. **Value:** detection, verification planning and legally grounded response.

Situation: Satellite analysis suggests a substantial increase in turbidity along a Bulgarian river reach. A settlement, a potentially vulnerable water intake and an industrial area lie downstream. This is a **hypothetical demonstration**, not a report of an actual pollution incident.

Typical workflow:

Step	System action	Specialist's action
1. Detection	Identify departure from historical baseline and create an incident record	Inspect primary observations
2. Confidence check	Account for clouds, seasonality, past observations and available measurements	Decide whether further investigation is justified
3. Spatial analysis	Identify water intakes and potentially affected downstream assets	Confirm which assets are relevant
4. Institutional competence	Suggest authorities and explain the jurisdictional basis	Confirm the correct organisations
5. Legal analysis	Display applicable rules, permit terms and procedural requirements	Verify sources and select an appropriate basis
6. Response	Draft sampling assignments and relevant notifications	Approve, send or reject
7. Outlook	If a valid model is available, estimate possible pollutant propagation	Interpret uncertainty and limitations
8. Closure	Track responsibilities, statuses, evidence and test results	Close the case or order additional measures

Example MVP incident card:

Field	Demonstration entry
Incident ID	WQ-2026-001
Overall status	Verification required
Signal	Elevated turbidity
Confidence	Moderate
Laboratory confirmation	None
Procedure status	Draft action plan

Recommendations for specialist review: (1) compare the observation with rainfall and river discharge; (2) check nearby monitoring stations; (3) assess whether sampling is necessary; (4) identify the competent authority and recipients; (5) document a reasoned decision on further steps.

The values and proposed actions above are illustrative. They do not establish that pollution has occurred or create any legal obligation.

Pain point addressed: less time between observing an anomaly and producing a defensible verification and interagency action plan.

What to show the jury: an event record, an identified recipient institution, a verifiable legal source, an action list and an approval trail—not merely a map.

Scenario 2. Flooding of an Industrial Zone

User: municipal emergency-management headquarters.

Situation: A hydrological forecast indicates possible inundation of an area containing factories, roads, warehouses and utility infrastructure.

System workflow:

1. Ingest the prospective flood footprint from a validated source.
2. Identify potentially affected roads, buildings, industrial sites and other facilities.
3. Classify impacts by timing and severity.
4. Refer to approved flood-risk management plans and applicable local procedures.
5. Identify agency and service responsibilities from verified national documentation.
6. Draft tasks for infrastructure checks, resource preparation and coordination with asset operators.
7. Record approvals and incoming updates.

The interface can show a baseline forecast, a worsening hydrological scenario, a scenario using additional flood-defence information, and the consequences of closing particular roads. These outcomes require a validated hydraulic or hydrological model. Generative AI must never invent flood depth.

Expected value: earlier mobilisation, faster consolidation of information and better coordinated decisions. FloodMapp and Copernicus already provide impact maps; our incremental value lies in country-specific procedures, responsibility and execution records.

Scenario 3. Drought and Drinking-Water Shortage

User: utility director or water-resource specialist.

Situation: Reservoir levels decline while demand remains high.

System workflow:

1. Collect reservoir levels, flows, precipitation and weather forecasts.
2. Flag a possible reduction in available water resources.
3. Retrieve registered reserve sources and relevant authorisations.
4. Map supply zones potentially subject to restrictions.
5. Suggest preparatory options and identify necessary approvals.
6. Log information requests, responses and operational decisions.

This is supported by a concrete Bulgarian policy example: in September 2025 the Ministry of Environment and Water urgently requested information on alternative water sources and permits. The need to compile these records quickly has therefore already arisen in practice.

Economic value: lower manual data-consolidation and decision-preparation costs, potentially with reduced consequences from supply interruptions.

Scenario 4. Insurer's Warning About Flood Exposure

User: insurance company risk analyst.

Situation: An emerging flood threat may affect multiple insured commercial properties.

System workflow:

1. Retrieve protected location data for insured assets under an authorised integration.
2. Overlay licensed or otherwise permitted flood-risk information.
3. Identify assets within the potential hazard footprint.
4. Estimate preliminary exposure of the portfolio.
5. Suggest alerts for appropriate employees and, where the insurer's procedures permit, policyholders.
6. Document preventative activities and follow-up.

Illustrative indicator	Value
Insured properties analysed	2,000
Properties within a potential inundation footprint	75
Properties needing priority review	18
Value of property potentially protected	Depends on insurer data
Expected insured loss	Must be calculated by a separate insurance model

The presence of an asset inside a forecast flood zone does not make an insured loss inevitable. Contract coverage, deductibles, asset characteristics and existing protective measures must be analysed independently. Since Fathom already provides sophisticated flood-risk analytics, our defensible role is to coordinate precautionary action using those models as inputs.

Scenario 5. Accidental Discharge at an Industrial Facility

User: environmental engineer or EHS manager.

Situation: An internal sensor or operator reports wastewater measurements outside normal process parameters, with a risk of breaching permitted operating conditions.

System workflow:

1. Log the signal and measurement source.
2. Compare it with the current, site-specific permit conditions.
3. Evaluate whether an applicable threshold is actually met, including measurement-method requirements.
4. Display measures provided for in the site's incident-response plan.
5. Identify any resulting duty to inform a competent authority.
6. Coordinate containment measures and further investigation.
7. Produce a traceable incident record for audit and management review.

Legal basis: Article 7 of Directive 2010/75/EU on industrial emissions, as amended, provides for measures where incidents or accidents significantly affect human health or the environment, including notification of the competent authority and efforts to limit impacts. The specific duties depend on facility type, permit conditions and national law.

Economic value: faster identification of a proper course of action, earlier containment, fewer administrative errors, better documentation and possible loss reduction from timely intervention.

6.1. Which Scenario Should Drive the MVP?

Criterion	River pollution	Flood	Drought	Insurance	Industrial incident
Fit with core concept	Very high	High	High	Medium	Very high
Demonstrating legal logic	Very suitable	Suitable	Suitable	Limited	Very suitable
Open-data availability	Medium	High	Medium	Limited	Low without partner
Competitive pressure	High	Very high	Medium	Very high	Medium
Credible MVP feasibility	High	High	Medium	Medium	High with data access
Recommended priority	1	2	3	5	4

Recommendation: build the first end-to-end demonstration around suspected river pollution. This directly reflects expertise in environmental inspection, administrative procedure and legal compliance.

7. Commercial Model and Economic Efficiency

This is the second most important area after competitor analysis. For a jury or prospective investor, we must answer two different questions: **Why is the software economically useful to the customer?** and **Why would that customer pay our company?** The second question is harder. Even a useful system may never receive public funding without an approved procurement or project.

7.1. Where Financial and Operational Value Comes From

Value source	Measurement method	Primary beneficiaries
Staff-time savings	Hours spent on each decision or report	Authorities, utilities and businesses
Fewer operational delays	Time from alert to approved action	All operational users
Reduced expected losses	Change in probability or severity of damage	Businesses, insurers and municipalities
Better documentation	Share of complete, accurate and traceable decisions	Authorities and regulated companies
Better-targeted monitoring	Need for repeat measurements and site visits	Inspectorates, laboratories, utilities
More efficient coordination	Number of hand-offs and time waiting for responses	Interagency teams

Any claim that the system reduces environmental or financial damage is currently a **hypothesis**. It must be demonstrated in a pilot or through defensible modelling. We cannot attribute the entire value of allegedly avoided damage to our platform without evidence.

7.2. Candidate Revenue Models

Model	Buyer	How it works	Main commercial risk

Model	Buyer	How it works	Main commercial risk
Annual SaaS subscription	Utility or enterprise	Fee based on capabilities, users and monitored sites	Unproven recurrent usage
Paid pilot	Company, utility or publicly funded project	One geographic area and limited procedures for 2–3 months	Implementation costs may be high
Enterprise licence	Large operator	Multi-department deployment with proprietary data and integrations	Long sales cycle
Public-sector programme	Ministry, directorate or municipality	Procurement or project-financed deployment	Budget and procurement constraints
API / analytics modules	Insurer or GIS integrator	Charging for access and processed events or assets	Competition from incumbent data providers
Methodological support	Any regulated operator	Procedure configuration and legal knowledge-base updates	Labour-intensive expert work

7.3. Illustrative Price Hypotheses

These figures are not verified competitor prices or established willingness to pay. They are only starting points for customer interviews.

Segment	Hypothetical annual price
Small utility or local operator	€5,000–€12,000
Industrial enterprise	€12,000–€30,000
Large water utility	€12,000–€30,000
Municipality or regional organisation	€10,000–€40,000
Insurance company with API and analytics	€20,000–€60,000
Multi-agency national system	Individually scoped deployment and support

The annual subscription must be separated from initial configuration, closed-source data connections, integration with existing IT, verification of national legal rules, commercial third-party data licences and user training.

7.4. Financial Illustration for a River Basin Directorate

Suppose one unit devotes **120 staff-hours per month** to examining alerts, preparing documentation, researching legal grounds and coordinating responses. These are modelling assumptions, not actual Bulgarian agency statistics.

Assumption or result	Amount
Manual workload	120 hours per month
Fully loaded hourly cost	€35
Potential effort reduction	35%

Assumption or result	Amount
Annual value of time released	120 × €35 × 35% × 12 = **€17,640**
Annual subscription	€12,000
One-off implementation	€6,000
Annualised cost over three years	€12,000 + €6,000 / 3 = **€14,000**
Hypothetical annual net benefit	**€3,640**
Benefit-to-cost ratio	Approximately **1.26**

This demonstrates the calculation method; it does not establish that any directorate would achieve these savings. Freed staff capacity is an operational benefit, not necessarily an equivalent cash saving.

7.5. Financial Illustration for an Industrial Enterprise

The business will pay if the expected value exceeds cost after considering implementation and operational risk. Consider the following assumptions:

- Staff time used for environmental reviews, documentation and procedural control: **100 hours per month**.
- Fully loaded labour cost: **€40 per hour**.
- Reduction in the relevant workload: **30%**.
- Annual probability of a selected accident scenario: **30%**.
- Consequences if it occurs: **€120,000**.
- Hypothetical reduction in the incident's impact due to timely intervention: **20%**.
- Annual platform subscription: **€14,000**.
- Implementation cost: **€6,000**, allocated across three years.

Component	Calculation	Annual result
Staff-time savings	100 × €40 × 30% × 12	€14,400
Hypothetical reduction in expected accident losses	30% × €120,000 × 20%	€7,200
Total modelled benefit	€14,400 + €7,200	**€21,600**
Annualised costs	€14,000 + €6,000 / 3	**€16,000**
Modelled net benefit	€21,600 − €16,000	**€5,600**

The resulting benefit-to-cost ratio is **1.35**. However, the 20% reduction in incident consequences is **not an observed result**. It must not be presented as proven product performance. Justification requires real incident data, measured decision times and evidence that interventions actually mitigate the consequences.

For the first corporate pilot, measure staff time, response speed and procedural accuracy. Evaluate avoided losses only at a later stage.

7.6. Financial Illustration for an Insurer

The insurer has two distinct sources of possible benefit.

A. More efficient analysis: quicker identification of insured assets in the hazard area; less manual geospatial processing; faster prioritisation; improved coordination of preventive actions with policyholders.

B. Possible reduction in claims: this can occur only if alerts trigger real actions that demonstrably reduce insured losses.

An illustrative expected-benefit equation is:

E = N × P × L × R

where **N** is the number of comparable insured assets or risk units; **P** is the annual probability of the relevant event; **L** is average insured loss if it occurs; **R** is the fraction of that loss actually avoided through preventive measures.

For **500 properties**, **2%** annual event probability per property, **€20,000** average covered loss, and a hypothetical **10%** reduction in loss, the model gives:

E = 500 × 0.02 × €20,000 × 0.10 = €20,000 per year.

This is a methodological illustration, not a real insurer forecast. It assumes comparability of assets and ignores many issues that a full actuarial model must address: correlated losses, policy terms, deductibles and protective measures already in place.

If our product costs **€25,000 annually**, this benefit alone is insufficient to establish a positive return. The insurance use case therefore needs more measurable benefits, a larger suitable portfolio or a different pricing structure. We should never casually promise a "20% reduction in claims."

7.7. Special Considerations in Bulgaria's Insurance Market

EEA estimates indicate that less than **3%** of historical economic losses from the relevant extreme weather and climate events in Bulgaria were privately insured over 1980–2024. That figure refers to the insured share of total losses, not penetration of any specific insurance policy.

It suggests long-term room for climate-risk insurance growth, but also potentially limits the immediate pool of commercial purchasers. I would not make insurers Bulgaria's only initial customer segment.

7.8. Simplified Economics of the Start-up

Measure	Model assumption
Average annual customer revenue	€18,000
Annual variable cost per customer	€3,000
Annual contribution per customer	€15,000
Annual fixed company cost	€120,000
Customers required to cover the modelled costs	**8**

Break-even customers = €120,000 / (€18,000 − €3,000) = 8.

With eight customers, annual revenue would be €144,000 and total modelled costs would also be €144,000. This model excludes material uncertainties including lengthy government sales processes, commercial-data subscriptions, delayed payments, churn, taxes, and the extra cost of adapting the legal layer to new countries. Eight customers are an arithmetic break-even point, not a forecast.

7.9. My Recommended Business Model

Phase 1 — Paid, focused pilot: one territory, one incident scenario, a defined set of inputs and measurable time-saving metrics; charge separately for setup and support.

Phase 2 — Recurring product: annual subscription, licences for additional territories or facilities, IT integrations and regular updates of the verified legal knowledge base.

Phase 3 — Expansion: insurance, flood and drought modules; entry into other EU markets; partnerships with monitoring and hydrological-data companies.

8. Future Demand: EU Regulation, Climate Risk and Technology

European water management is likely to face significant changes over the next five to ten years. The product should be prepared for this direction while avoiding unfounded claims that EU law mandates the purchase of any particular software.

8.1. Principal EU Instruments

EU legal instrument	What it regulates	Relevance to Water Crisis Decision Navigator
Directive **2000/60/EC**, Water Framework Directive	Water-body management, environmental objectives, monitoring and river-basin planning	Link observations with water-body status, management plans and institutions
Directive **2007/60/EC**, Floods Directive	Flood-risk assessment and management	Integrate approved hazard maps, flood-risk plans and response procedures
Directive **2004/35/EC**, Environmental Liability Directive	Prevention and remediation of environmental damage	Support identification of applicable preventive or remedial measures
Directive **2010/75/EU**, as amended, Industrial Emissions Directive	Industrial permits, pollution prevention, incidents and emissions	Review permit terms and prepare action proposals for regulated installations
Directive **2012/18/EU**, Seveso III	Major-accident hazards involving dangerous substances	Link to emergency plans for high-risk installations
Directive **2020/2184**, Drinking Water Directive	Drinking-water quality and risk-based water-supply safety	Water-intake risk assessment and utility coordination
Directive **(EU) 2024/3019**, Urban Wastewater Treatment Directive	Collection, treatment and oversight of urban wastewater	Potential future modules for wastewater operators
Directive **(EU) 2026/805**	Updates rules on surface-water and groundwater pollutants	Adapt pollutant parameters and national operating procedures
Regulation **(EU) 2024/1689**, AI Act	Governance of artificial-intelligence systems	Determine applicable risk category, oversight and traceability
Directive **(EU) 2022/2555**, NIS2	Cybersecurity for many essential and important entities, including parts of the water sector	Secure deployment, integration and access management

Important legal qualification: EU directives are implemented through national law. The duties of any particular organisation also depend on its legal status, permit conditions, the event involved and local procedures. Loading EU directive texts into an AI model does not by itself create an authoritative Bulgarian response manual.

8.2. A Major Legislative Change in 2026: Directive (EU) 2026/805

According to the underlying research, Directive **(EU) 2026/805** entered into force on **11 May 2026**, amending three parts of EU water-protection legislation: the Water Framework Directive, environmental quality standards for surface waters and groundwater protection. The amendments address, among other topics:

- Certain PFAS compounds.
- Pharmaceuticals and pesticides.
- Microplastics.
- Indicators of antimicrobial resistance.
- Assessment of combined pollutant effects.
- Revised approaches to monitoring.

Commercial implication: interpreting monitoring results and identifying the requirements that apply to a particular site or water body will become more demanding.

A future module called **Water Regulation Intelligence** should (1) follow legislative amendments; (2) determine relevant commencement dates; (3) account for national transposition; (4) match duties to water-body type; (5) flag additional monitoring parameters; (6) alert organisations to procedural revisions; and (7) link to primary legal texts and specific versions.

The source analysis identifies **21 December 2027** as the deadline for Member States to adopt transposition measures. Individual provisions may have separate transition dates. National transposition does not mean that every environmental standard has exactly the same effective date. Primary source: https://eur-lex.europa.eu/legal-content/en/TXT/?uri=CELEX:32026L0805.

8.3. New Drinking-Water Requirements

On **12 January 2026**, new EU-wide requirements for monitoring PFAS in drinking water began to apply under the recast Drinking Water Directive. The European Commission describes harmonised monitoring and information obligations relating to results, exceedances and incidents.

This may create demand for combined laboratory data, checking applicable thresholds, identifying affected supply systems, preparing reports and tracking required action.

Technical boundary: satellite imagery must not be advertised as a stand-alone measurement method for PFAS. Relevant laboratory testing is required. Commission source: https://environment.ec.europa.eu/news/new-eu-rules-limit-pfas-drinking-water-2026-01-12_en.

8.4. Floods and River Basin Management

Directive **2007/60/EC** requires flood-risk assessment, hazard and risk mapping, and flood-risk management plans. These documents are updated periodically; the current planning cycle referred to in the underlying analysis covers **2022–2027**.

The platform should use existing approved maps and plans. Outputs from experimental algorithms must not silently replace them.

8.5. Environmental Liability and Industrial Safety

Directive **2004/35/EC** is especially relevant to regulated businesses. Article 5 addresses preventive measures where there is an imminent threat of environmental damage, including notification of competent authorities in applicable cases. Article 6 addresses action after environmental damage has occurred.

An industrial response module might generate a verified checklist, draft internal notification, relevant permit conditions, a record of preventive measures, a dated chronology and a draft incident report. It must **not** purport to establish a company's legal liability automatically.

8.6. EU AI Act: Opportunity and Constraint

Because the system could eventually be used in critical infrastructure, the possible application of the EU **AI Act** deserves early examination. Certain AI systems used as safety components in the management and operation of critical infrastructure, including water supply, may be classified as **high risk**. The classification depends on the product's specific intended purpose and technical role; using AI in an environmental organisation does not automatically make the system high risk.

The regulatory framework has category-specific compliance timetables, including provisions associated with **December 2027** for certain Annex III cases as described in the underlying source analysis. Detailed classification and timing need to be checked against the law in force when a commercial use case is defined.

From the start, design for human approval of significant actions, a recorded source for every material conclusion, versioned legal references, explicit uncertainty, correction and challenge mechanisms, a decision log and role-based access rights. These safeguards will matter to public buyers and industrial customers.

8.7. Cybersecurity and NIS2

NIS2 applies to certain water-supply and wastewater entities and other critical sectors where its criteria are met. A future enterprise system should address confidential operational data, permission control, encryption, audit logs, backups, secure exchange with third-party systems and cyber-incident procedures.

A safe demonstrator architecture is sufficient for the hackathon MVP. Production-grade compliance and assurance will become a material cost of commercial deployment. Source: https://eur-lex.europa.eu/eli/dir/2022/2555/oj.

8.8. European Water Resilience Policy

The **EU Water Resilience Strategy 2025** supports more efficient water use, resilient management, innovation and digitalisation. This offers a policy and investment backdrop, not a legal duty to buy Water Crisis Decision Navigator.

The European Commission's **Water Resilience Investment Accelerator**, announced on **14 October 2026**, is particularly worth examining. It envisages support for **20 innovative pilot projects** and investment-preparation assistance, including through **Green Assist**. The Commission's original announcement indicated an opportunity to seek advisory support until late October 2026. Such advisory support is not a guaranteed cash grant, and eligibility must be checked separately.

Given the source report's reference date of **25 October 2026**, the team should examine this option promptly if it intends to continue the project after the hackathon. Source: https://environment.ec.europa.eu/news/commission-presents-water-resilience-investment-accelerator-2026-10-14_en.

8.9. Technological Trends Shaping Demand

Technology	Expected trajectory	Opportunity for our product
Satellite monitoring	More accessible repeat observations	Regular detection of change
IoT and field sensors	More frequent operational measurements	Corroborate signals across sources
Hydrological models	Improved spatial resolution and forecasting	Analyse hazard development
GIS platforms	Tighter institutional data integration	Connect hazards to specific assets and places

Technology	Expected trajectory	Opportunity for our product
Generative AI	Faster handling of documents and information retrieval	Draft evidence-linked decisions for expert review
Business-process automation	Growing demand for transparent digital procedure	Assign tasks and monitor execution
Digital regulation	More formal requirements for data and reporting	Maintain a version-controlled regulatory knowledge base

The most promising combination is geospatial intelligence + verified regulation + managed operational workflows.

9. MVP Validation and Proof of Demand

This is central to **Stavri Nikolov's** recommendations. Winning a hackathon and demonstrating genuine market demand are not equivalent. An MVP must test not only whether the technology can be built, but also whether customers experience a problem serious enough to pay to solve.

9.1. Hypotheses to Test

Hypothesis	Validation method	Success evidence
Coordination is a significant problem	Interviews and direct observation	Recurring, specific process failures described by staff
The product accelerates work	Comparable task tests	Measured reduction in processing time
Legal references help users	Expert examination	Correct, applicable and versioned legal sources
Responsible bodies are assigned correctly	Walkthrough of official procedures	No critical jurisdictional mistakes
Users trust the explanation	Usability evaluation	Users understand both the basis and limitations
Commercial demand exists	Discussion with an actual budget holder	Serious consideration of a defined paid pilot
Deployment is feasible	IT and cybersecurity review	No insurmountable integration barriers

9.2. A Realistic Minimum Viable Product

Geographic scope: one water body or a short river reach in Bulgaria.

Hazard: turbidity or another observable parameter that can be demonstrated reliably with available datasets.

Inputs: one open geospatial-data source and one sample of historical, field-measurement or clearly labelled demonstration laboratory data.

Participants: environmental analyst, authorised technical specialist, representative of the responsible inspectorate, and manager approving the decision.

Legal component: at least **3–5 verified rules or procedural steps** selected for the specific geographic jurisdiction and type of event.

Output: threat record, proposed action plan, legal citations, named task owners, approval status and incident timeline.

9.3. Example MVP Technical Flow

1. Import time-stamped environmental data from a defined source.
2. Identify and visualise an apparent anomaly while keeping uncertainty visible.
3. Combine the event's location with a curated map of sensitive receptors and authority jurisdictions.
4. Retrieve exact reviewed procedures and legal references appropriate to that place and event category.
5. Create a proposed check-and-notify workflow for a specialist to approve.
6. Record decisions, assignments, supporting evidence and completion status.

This is a **decision-support demonstration**, not an autonomous crisis-control platform.

9.4. Programme of Potential-Customer Interviews

I recommend **8–10 interviews** distributed across roles as follows:

Interview group	Number	Main purpose
River Basin Directorate professionals	2	Validate jurisdiction and procedures
Regional environmental specialists	2	Reconstruct actual response workflows
Water utilities	2	Find commercially important recurring tasks
Industrial enterprises	1–2	Examine incident procedures and acceptable pricing
Insurance company	1	Understand unmet analytics and prevention needs
IT specialist or procurement officer	1	Assess deployment and budget pathway

This is a proposed research plan; the interviews have **not yet taken place**.

9.5. Interview Questions

In line with Nikolov's advice, begin with past events and real work rather than abstract opinions about the concept.

Current Work

1. Tell me about the most recent case when a possible water-quality threat was identified.
2. Where did the initial warning come from?
3. Who checked whether the information was reliable?
4. Which documents and software systems were used?
5. How much time elapsed between the alert and the first decision?

Response Organisation

6. How was the responsible public authority identified?
7. Which specialists participated?
8. How was information exchanged?
9. What documentation had to be prepared?

10. At what stage was the longest delay?

Legal and Procedural Issues

11. How do employees find the latest applicable legal requirements?
12. Which site-specific permit conditions must be considered?
13. Which decisions require mandatory review or authorisation?
14. Where is the history of actions and notifications stored?
15. Which errors would have the most serious consequences?

Economics and Workload

16. How many comparable incidents occur each year?
17. How much working time does one case require?
18. What costs arise from manual coordination?
19. Does the organisation already pay for information systems?
20. How is a decision to purchase new software assessed?

Procurement

21. Who may propose introducing a new system?
22. Who authorises the expenditure?
23. What procurement procedure applies?
24. Could the organisation commission a limited paid pilot?
25. What pilot results would justify considering a longer-term contract?

9.6. Observe the Actual Working Process

I support Nikolov's recommendation to observe work inside a participating organisation, if it gives permission. In particular, examine how many systems and tabs an officer opens; how information is transferred between documents; how long legal research takes; who has to be called; how receipt of information is confirmed; who authorises a decision; and which steps are repeated.

Observation can reveal friction that professionals no longer mention because it has become routine. It must not involve unauthorised access to operational data. Anonymised records or a controlled demonstration are sufficient for an early-stage start-up.

9.7. How to Measure MVP Performance

The strongest design is a comparison of equivalent tasks **without the platform** and **with the platform**.

Test assignment: investigate an alert concerning possible water contamination and prepare a reasoned draft plan for next steps.

Indicator	Conventional process	MVP-assisted process	What is measured
Time to analyse an alert	Record actual time	Record actual time	Speed
Time to identify competent authority	Record actual time	Record actual time	Speed
Time to find legal grounds	Record actual time	Record actual time	Speed

Indicator	Conventional process	MVP-assisted process	What is measured
Accuracy of legal references	Reviewed by expert	Reviewed by expert	Quality
Correctness of notification recipients	Reviewed by expert	Reviewed by expert	Quality
Completeness of required actions	Reviewed by expert	Reviewed by expert	Quality
User's understanding of results	Questionnaire	Questionnaire	Usability
Number of critical errors	Count	Count	Safety

Suggested preliminary success criteria:

- At least **30% less time** needed to prepare the initial justified decision.
- **No critical errors** in identifying the competent authority.
- All legal citations in the approved demonstration set lead to the correct source.
- Users understand which findings are evidenced and which remain hypotheses.
- Every significant action is subject to appropriate human approval.

These are **targets for pilot evaluation**, not outcomes that the product has already achieved.

9.8. How to Test Willingness to Pay

"Do you like our idea?" provides almost no reliable evidence of demand. A more useful question is:

"If a two-month pilot could reduce decision-preparation time by 30% while maintaining legal correctness, would you consider a paid implementation? Who would need to authorise the pilot, and from which budget line could it be financed?"

Interest level	Observed signal	Commercial significance
1	User says the idea is interesting	Low
2	User agrees to a demonstration	Initial curiosity
3	Organisation commits staff time and relevant data	Meaningful signal
4	Authorised representative discusses budget, terms and timing of a paid pilot	Strong signal

Even level 4 is not a signed contract.

Overall Expert Assessment

Strengths

- A clear societal, environmental and economic problem.
- Strong alignment with European water-policy priorities.
- Opportunity to draw on existing satellite and hydrological data.
- Applicability across several customer categories.
- Potentially high value from the institutional and legal workflow layer.

- Modular expansion across Member States and threat types.

Your experience in public environmental administration, inspections, impact assessment and external audits is particularly relevant to the procedural components: competence matrices, verification scenarios, formal records and decision checkpoints.

Main Risks and Mitigations

Risk	Severity	Recommended mitigation
Strong existing vendors	High	Integration and narrow specialisation
Complexity of national law	High	Start with one country and one procedure
Satellite data cannot measure every pollutant accurately	Critical	Separate suspected anomalies from confirmed findings
Legally incorrect AI recommendations	Critical	Expert-verified sources, human approval and version control
Slow government procurement	High	Test corporate buyers in parallel
Unsupported savings claims	High	Quantified pilots with documented assumptions
Attempting pollution, floods and droughts simultaneously	High	Build one complete MVP scenario
Dependence on third-party models and datasets	Medium	Account for licences and alternative sources

What to Give Francesco First

With limited time before preparing the presentation, I would prioritise the following five items:

Priority	Deliverable	How Francesco can use it
1	Defensible competitor comparison	Position the product accurately without claiming there are no competitors
2	Detailed river-pollution user story	Foundation for the MVP demonstration
3	Customer, user and budget-owner matrix	Explain the intended buyers and target market
4	Three economic illustrations for public authorities, industry and insurers	Preliminary business justification
5	Eight-to-ten-interview plan and MVP test design	Demonstrate a credible demand-validation strategy

Final Recommendation and Positioning

Water Crisis Decision Navigator is a decision-support platform for environmental authorities and water-sector organisations. It links monitoring data to verified legal requirements, the powers of competent authorities and documented response procedures.

First product: detection and documented verification of suspected river pollution in Bulgaria.

First target customer: a River Basin Directorate or a water-sector operator.

Principal measurable result: reduced time needed to prepare justified decisions without sacrificing legal accuracy.

Later modules: floods, droughts, industrial environmental safety and insurance risk.

I consider this the most realistic route to post-hackathon development. It builds on existing environmental datasets, leverages practical public-administration expertise and makes it possible to test real willingness to pay before attempting to create a costly universal forecasting platform.

The defining next question is not merely whether the technology is impressive, but **whether real organisations will pay to save time and improve the quality, legal defensibility and traceability of decisions**. The first interviews and pilot should test this before the nine-slide pitch is finalised.

Primary Sources and Further Reading

1. European Environment Agency — economic losses from extreme weather and climate events: https://www.eea.europa.eu/en/analysis/indicators/economic-losses-from-climate-related

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3. Bulgarian Ministry of Environment and Water — combined flood-risk maps: https://www.moew.government.bg/en/minister-manol-genov-commissioned-the-preparation-of-combined-maps-of-flood-risk-areas/

4. Copernicus Emergency Management Service: https://emergency.copernicus.eu/about/

5. FloodMapp: https://www.floodmapp.com/

6. Fathom — insurance applications: https://www.fathom.global/sector/insurance/

7. EOMAP eoapp AQUA: https://eomap.com/start-download/eoapp-aqua-leaflet

8. ArcGIS Workflow Manager: https://doc.esri.com/en/arcgis-workflow-manager/latest/help/an-introduction-to-arcgis-workflow-manager.html

9. Directive (EU) 2026/805: https://eur-lex.europa.eu/legal-content/en/TXT/?uri=CELEX:32026L0805

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15. EU AI Act, consolidated text: https://eur-lex.europa.eu/legal-content/EN/TXT/HTML/?uri=CELEX:02024R1689-20260727

16. NIS2 Directive: https://eur-lex.europa.eu/eli/dir/2022/2555/oj

English-language translation and editorial adaptation of the expert report dated 25 October 2026. Economic scenarios are illustrative assumptions, not verified financial results or price quotations. Public legal and competitor claims should be checked against their primary sources before a production deployment or investment decision.